Primary reason recorded for efficacy failure in the 27 drug classes whose target engagement was confirmed in patients

Target hypothesis: the target and the direction of the intended effect — n = 13

- ◆ Not causal: output moved, disease did not (9)
  - anti-IL-13 · asthma
  - anti-IL-33 · atopic eczema
  - p38 inhibitors · COPD
  - HCAR2 agonists · lipids
  - HDAC inhibitors · MDS
  - anti-IGF-1R · NSCLC
  - PDE10A inhibitors · SCZ
  - GPR119 agonists · T2D
  - 11β-HSD1 inh. · T2D
- ◆ Causal at another disease stage (1)
  - BACE1 inhibitors · AD
- ◆ Causal only in a subgroup (0)
  - *none*
- Causal but compensated or bypassed (1)
  - anti-EGFR (new) · CRC
- Causal, effect too small (2)
  - pan-PI3K inhibitors · breast
  - MEK inhibitors · pancreas

Modulation: the drug, its dose, or the site of engagement — n = 1

- Insufficient degree or duration of engagement (1)
  - anti-GM-CSF · RA

Trial: endpoint, population, or treatment context — n = 4

- Population or placebo response (3)
  - allergen peptides · allergy
  - allergen IT · rhinitis
  - anti-CGRP · migraine
- Combination or comparator context (1)
  - DP2 antagonists · asthma

Not stated in the record — n = 9

- Not stated in the record (9)
  - DP2 antagonists · rhinitis
  - H3 antagonists · AD
  - 5-HT6 antagonists · AD
  - HSP90 inhibitors · breast
  - PI3K/mTOR inh. · breast
  - NRTIs (new) · HIV
  - BTK inhibitors · RA
  - CDK4/6 inhibitors · breast
  - proteasome inh. · myeloma

class never approved in this disease

class approved in this disease; the contradicted entry concerns a stage, a population, or one member

◆ category present in the current manuscript draft; the remaining categories are added here